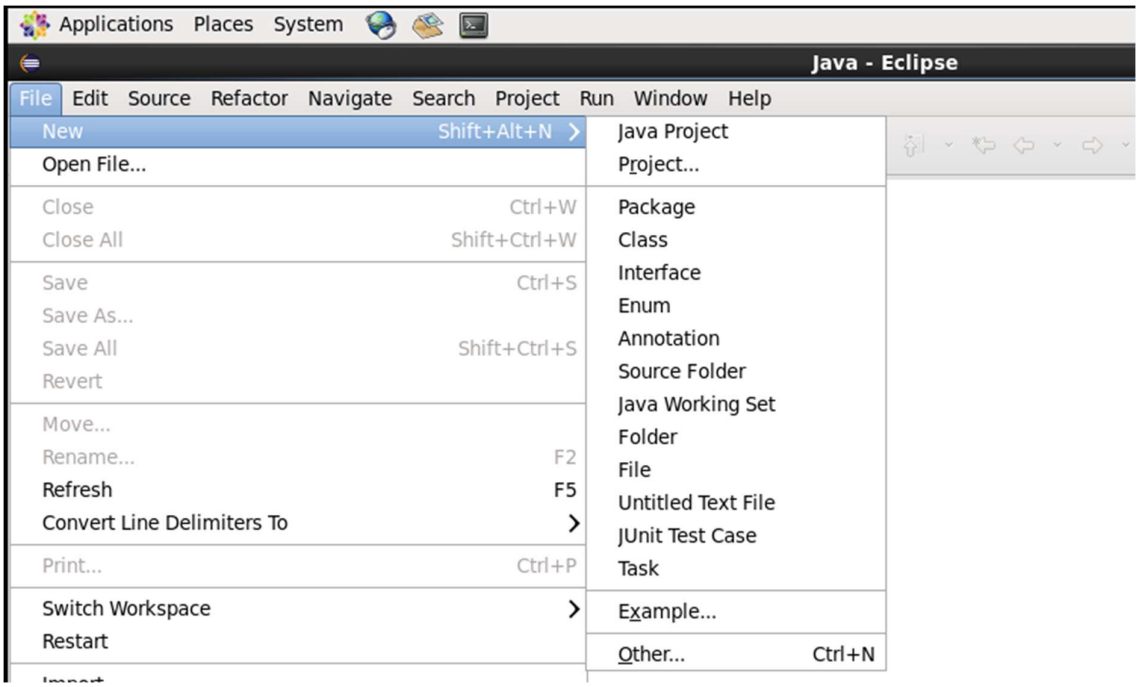
# 3. Map Reduce application for word counting on Hadoop cluster

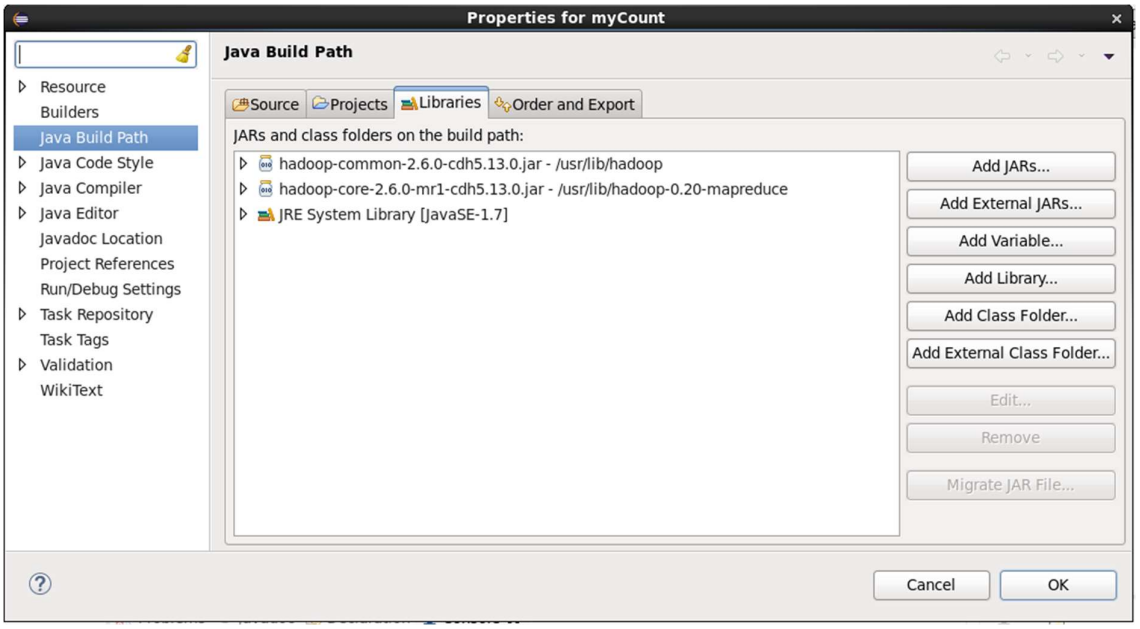
**STEPS:**

**Step1:** Open the Eclipse application.

**Step2:** Create a new Java project name: WordCount




**Step3:** Add External Jar Files



**Step4:** Add **/usr/lib/Hadoop** all jar files

**Step5:** Add **/usr/lib/Hadoop/client** all jar files

**Step6:** Create java class name WordCount.java

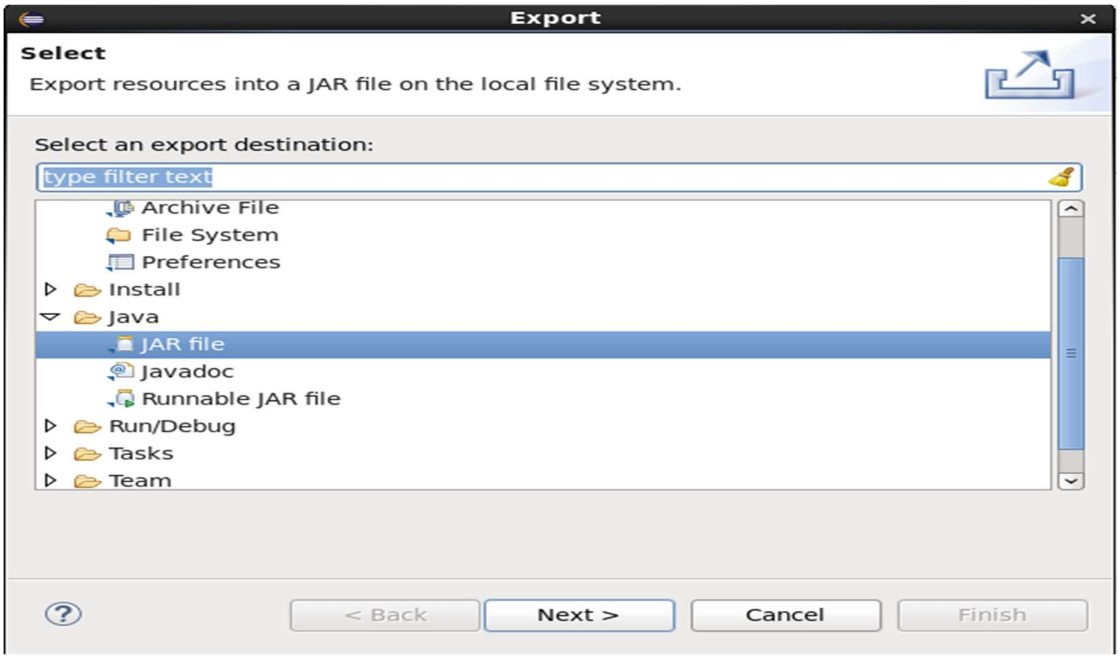
**Step7:** Copy-paste the code of WordCount on WordCount.java file

**Source Code for WordCount:**

```
import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {

  public static class TokenizerMapper
      extends Mapper<Object, Text, Text, IntWritable>{

    private final static IntWritable one = new IntWritable(1);
    private Text word = new Text();

    public void map(Object key, Text value, Context context
                ) throws IOException, InterruptedException {
      StringTokenizer itr = new StringTokenizer(value.toString());
      while (itr.hasMoreTokens()) {
        word.set(itr.nextToken());
        context.write(word, one);
      }
    }
  }

  public static class IntSumReducer
      extends Reducer<Text,IntWritable,Text,IntWritable> {
    private IntWritable result = new IntWritable();

    public void reduce(Text key, Iterable<IntWritable> values,
                    Context context
                    ) throws IOException, InterruptedException {
      int sum = 0;
      for (IntWritable val : values) {
        sum += val.get();
      }
```

```
      result.set(sum);
      context.write(key, result);
    }
  }

  public static void main(String[] args) throws Exception {
    Configuration conf = new Configuration();
    Job job = Job.getInstance(conf, "word count");
    job.setJarByClass(WordCount.class);
    job.setMapperClass(TokenizerMapper.class);
    job.setCombinerClass(IntSumReducer.class);
    job.setReducerClass(IntSumReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);
    FileInputFormat.addInputPath(job, new Path(args[0]));
    FileOutputFormat.setOutputPath(job, new Path(args[1]));
    System.exit(job.waitForCompletion(true) ? 0 : 1);
  }
}
```

**Step8:** Export the Jar file name WordCount.jar to /home/cloudera. Right Click on Project-> Click on Export-> Select export destination as Jar File-> Name the jar File (WordCount.jar) -> Click on next -> at last Click on Finish.




**Step9:** Close the Eclipse.

**Step10:** Open Terminal, change directory to /home/Cloudera.

**Step11:** Create a text file name Processfile1.txt and write text in it.

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ pwd
/home/cloudera
[cloudera@quickstart ~]$ cat > /home/cloudera/Processfile1.txt
This is Manu
This is xyz
This is BigData
[cloudera@quickstart ~]$ cat /home/cloudera/Processfile1.txt
This is Manu
This is xyz
This is BigData
[cloudera@quickstart ~]$ hdfs dfs -ls /
```

**Step12:** Save it with CTRL+D.

**Step13:** Now create a input directory in hdfs.

**Syntax:** hdfs dfs -mkdir /input_dir

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ hdfs dfs -mkdir /input_dir
[cloudera@quickstart ~]$ hdfs dfs -ls /
Found 12 items
drwxrwxrwx   - hdfs     supergroup          0 2017-10-23 10:29 /benchmarks
-rw-r--r--   1 cloudera supergroup        110 2022-03-20 23:25 /bigdata
drwxr-xr-x   - hbase    supergroup          0 2022-04-02 22:08 /hbase
drwxr-xr-x   - cloudera supergroup          0 2022-04-02 22:39 /input_dir
-rw-r--r--   4 cloudera supergroup         12 2022-03-26 22:00 /mydir
-rw-r--r--   1 cloudera supergroup          0 2022-03-26 22:11 /mydir1
-rw-r--r--   1 cloudera supergroup         32 2022-03-26 22:20 /mydir3
-rw-r--r--   1 cloudera supergroup         12 2022-03-27 07:37 /mydir_copied
drwxr-xr-x   - solr     solr                0 2017-10-23 10:32 /solr
drwxrwxrwt   - hdfs     supergroup          0 2022-03-20 23:51 /tmp
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 10:31 /user
drwxr-xr-x   - hdfs     supergroup          0 2017-10-23 10:31 /var
```

**Step14:** Now put this file in Hadoop using

**Syntax:** hdfs dfs -put /home/cloudera/Processfile1.txt /input_dir/

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ hdfs dfs -put /home/cloudera/Processfile1.txt /input_dir/
[cloudera@quickstart ~]$ hdfs dfs -ls /input_dir
Found 1 items
-rw-r--r--   1 cloudera supergroup         41 2022-04-02 22:44 /input_dir/Processfile1.txt
[cloudera@quickstart ~]$ hdfs dfs -cat /input_dir/Processfile1.txt/
This is Manu
This is xyz
This is BigData
```

**Step15:** Now to run the program write

**Syntax:** Hadoop jar home/cloudera/WordCount.jar WordCount /input_dir/Processfile1.txt/out11

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ hadoop jar /home/cloudera/WordCount.jar WordCount /input_dir/Processfile1.txt /out11
22/04/02 22:52:41 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
22/04/02 22:52:44 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implemen
ol interface and execute your application with ToolRunner to remedy this.
22/04/02 22:52:45 INFO input.FileInputFormat: Total input paths to process : 1
22/04/02 22:52:46 INFO mapreduce.JobSubmitter: number of splits:1
22/04/02 22:52:47 WARN hdfs.DFSClient: Caught exception
java.lang.InterruptedException
        at java.lang.Object.wait(Native Method)
        at java.lang.Thread.join(Thread.java:1281)
        at java.lang.Thread.join(Thread.java:1355)
        at org.apache.hadoop.hdfs.DFSOutputStream$DataStreamer.closeResponder(DFSOutputStream.java:967)
```

**Step16:** To see the output write

**Synatx:** hdfs dfs -cat /out11/part-r-00000

```
cloudera@quickstart:~
File  Edit  View  Search  Terminal  Help
[cloudera@quickstart ~]$ hdfs dfs -ls /out11
Found 2 items
-rw-r--r--   1 cloudera supergroup          0 2022-04-02 22:54 /out11/_SUCCESS
-rw-r--r--   1 cloudera supergroup         35 2022-04-02 22:54 /out11/part-r-00000
[cloudera@quickstart ~]$ hdfs dfs -cat /out11/part-r-00000
BigData 1
Manu    1
This    3
is      3
xyz     1
```